

Problem 3.

- 1st person sg:
 - if the word contains at least one voiceless consonant, the first one becomes nasal+voiced ($\langle \mathbf{p} \rightarrow \mathbf{mb}, \mathbf{t} \rightarrow \mathbf{nd}, \mathbf{h} \rightarrow \mathbf{nz}, \mathbf{x} \rightarrow \mathbf{nj}, \mathbf{k} \rightarrow \mathbf{ng} \rangle$)
 - otherwise the whole word is nasalised ($\mathbf{-m}$)
- 2nd person sg:
 - if the word begins with a vowel different from $\mathbf{i}$ ($\mathbf{\acute{i}}, \mathbf{\hat{i}}$): $\mathbf{y-}$
 - otherwise the first vowel different from $\mathbf{i}$ ($\mathbf{\acute{i}}, \mathbf{\hat{i}}$) changes as follows:
 - * $\mathbf{a} \rightarrow \mathbf{e}, \mathbf{\acute{a}} \rightarrow \mathbf{\acute{e}}, \mathbf{\hat{a}} \rightarrow \mathbf{e\hat{a}}$;
 - * $\mathbf{o} \rightarrow \mathbf{e}, \mathbf{\acute{o}} \rightarrow \mathbf{\acute{e}}, \mathbf{\hat{o}} \rightarrow \mathbf{e\hat{o}}$;
 - * $\mathbf{\hat{u}} \rightarrow \mathbf{i\hat{u}}$;
 - * $\mathbf{e} \rightarrow \mathbf{i}, \mathbf{\acute{e}} \rightarrow \mathbf{\acute{i}}, \mathbf{\hat{e}} \rightarrow \mathbf{\hat{i}}$ (the same happens in the immediately following syllables).

(a)	1st person sg	2nd person sg		1st person sg	2nd person sg	
	<i>mbího</i>	₁ píhe	<i>to go</i>	<i>noínjoa</i>	₈ neíxoá	<i>to see it</i>
	₂ mbêyo	<i>pîyo</i>	<i>animal</i>	<i>vanénjo</i>	₉ venéxo	<i>to buy</i>
	<i>yênom</i>	₃ yîno	<i>wife</i>	<i>mómindi</i>	₁₀ mémiti	<i>to be tired</i>
	₄ ênom	<i>yêno</i>	<i>mother</i>	<i>ínzikaxovoku</i>	₁₁ ihikexovoku	<i>school</i>
	<i>ngásaxo</i>	₅ késaxo	<i>to feel cold</i>	₁₂ ônju	<i>yôxu</i>	<i>grandfather</i>
	<i>njérere</i>	₆ xíriri	<i>side</i>	₁₃ nêwom	<i>nîwo</i>	<i>nephew</i>
	<i>ndôko</i>	₇ teôko	<i>nape</i>	<i>ánzarana</i>	₁₄ yáharana	<i>hoe</i>

- (b) 1. 2nd person sg: in Portuguese loanwords $\mathbf{\acute{a}} \rightarrow \mathbf{e\acute{a}}$; in native words $\mathbf{\acute{a}} \rightarrow \mathbf{\acute{e}}, \mathbf{\hat{a}} \rightarrow \mathbf{e\hat{a}}$.
2. **lándana, ngápana.**